

Auto Shift V1 Case Study

This case study explains how Auto Shift V1 was designed and integrated into a production shift management system used by 9 stores and 130+ staff members.

Background

The original system allowed staff members to submit their available working times and allowed store managers to review submissions, manage staff, configure scheduling periods, set business hours, and export finalized shifts to Excel.

However, the final shift creation process was still mostly manual.

Managers had to check submitted availability, required staffing levels, opening and closing coverage, kitchen/hall positions, and store-specific business hours before creating the final schedule.

Problem

Manual shift creation becomes difficult when multiple operational rules need to be considered at the same time.

The main issues were:

- Managers had to manually check whether each time slot had enough staff
- Opening and closing coverage had to be checked separately
- Some stores required kitchen/hall staff separation
- Common staff could be assigned differently depending on manager judgment
- Existing Excel export behavior could not be broken
- Original staff availability submissions had to be preserved
- Managers must only access data from their own store

Because the system was already used in real store operations, the goal was not to immediately build a fully automatic scheduling algorithm.

The first goal was to create a safe draft-generation flow that managers could review and edit.

Goals

Auto Shift V1 was designed with the following goals:

- Generate an initial shift draft from submitted availability
- Allow managers to edit the generated draft before confirmation
- Preserve original staff availability data
- Store the generated draft and confirmed result separately
- Keep existing Excel export workflows working
- Create data that can be used later to analyze manager edits
- Minimize operational risk during deployment

Design

To avoid damaging production data, the system separates shift data into multiple stages.

- **availabilities**: original staff submissions
- **draftRows**: auto-generated shift draft
- **workingRows**: manager-edited temporary state in the UI
- **confirmedRows**: manager-confirmed final schedule
- **confirmed_shift_snapshots**: final snapshot saved at confirmation time

This design makes the workflow safer because the original submitted availability is never overwritten.

It also makes future analysis possible by comparing the auto-generated draft with the manager-confirmed result.

Auto Shift V1 Logic

Auto Shift V1 is a rule-based draft generation system.

It considers:

- Staff submitted availability
- Store business hours
- Required staff settings
- Opening coverage
- Closing coverage
- Kitchen / hall / common staff positions
- Shortage detection by time slot

The generated result is not treated as the final answer.

Instead, it is shown to the manager as an editable draft.

Operational Considerations

Because this system is used in real stores, stability was more important than adding complex optimization features quickly.

Important constraints included:

- Do not overwrite original staff availability data
- Do not break existing manual Excel export behavior
- Prevent managers from accessing other stores' data
- Keep privileged Supabase access on the server side only
- Store generated drafts and confirmed schedules separately
- Save confirmed snapshots for future analysis

Result

Auto Shift V1 introduced the following workflow:

1. Staff submit availability
2. The system generates an automatic draft

3. The manager reviews and edits the draft
4. The manager confirms the final schedule
5. The system saves a confirmed snapshot
6. The final schedule can be exported to Excel

This created the foundation for future improvements based on real manager edit data.

Future Work

The next step is to improve Auto Shift using the data collected from manager edits.

Planned improvements include:

- Add row-level identifiers for more accurate diff analysis
- Analyze differences between `draftRows` and `confirmedRows`
- Collect manager edit reasons
- Add scoring for generated drafts
- Support date-specific required staff overrides
- Improve Auto Shift V2 using real operation data